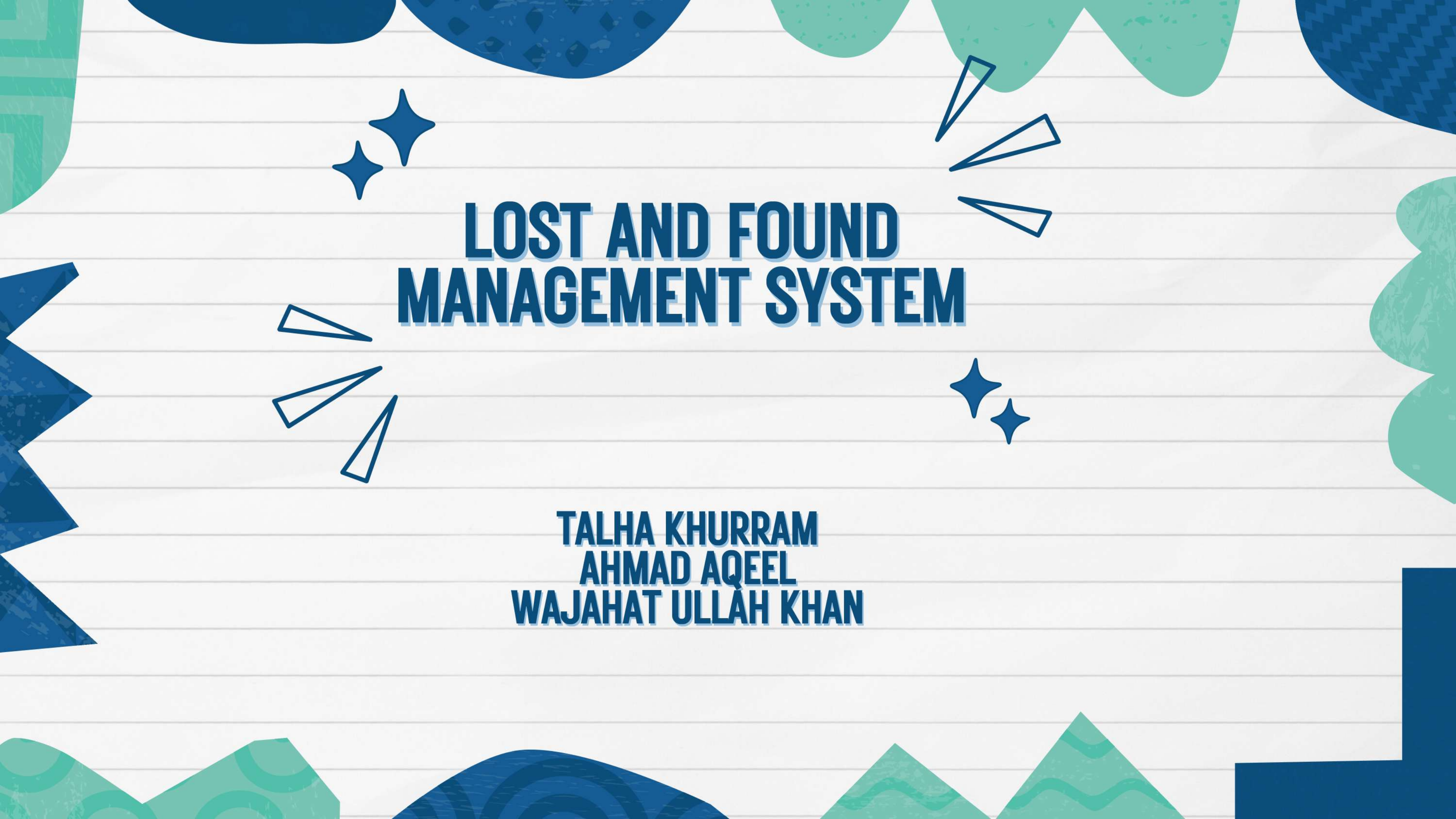
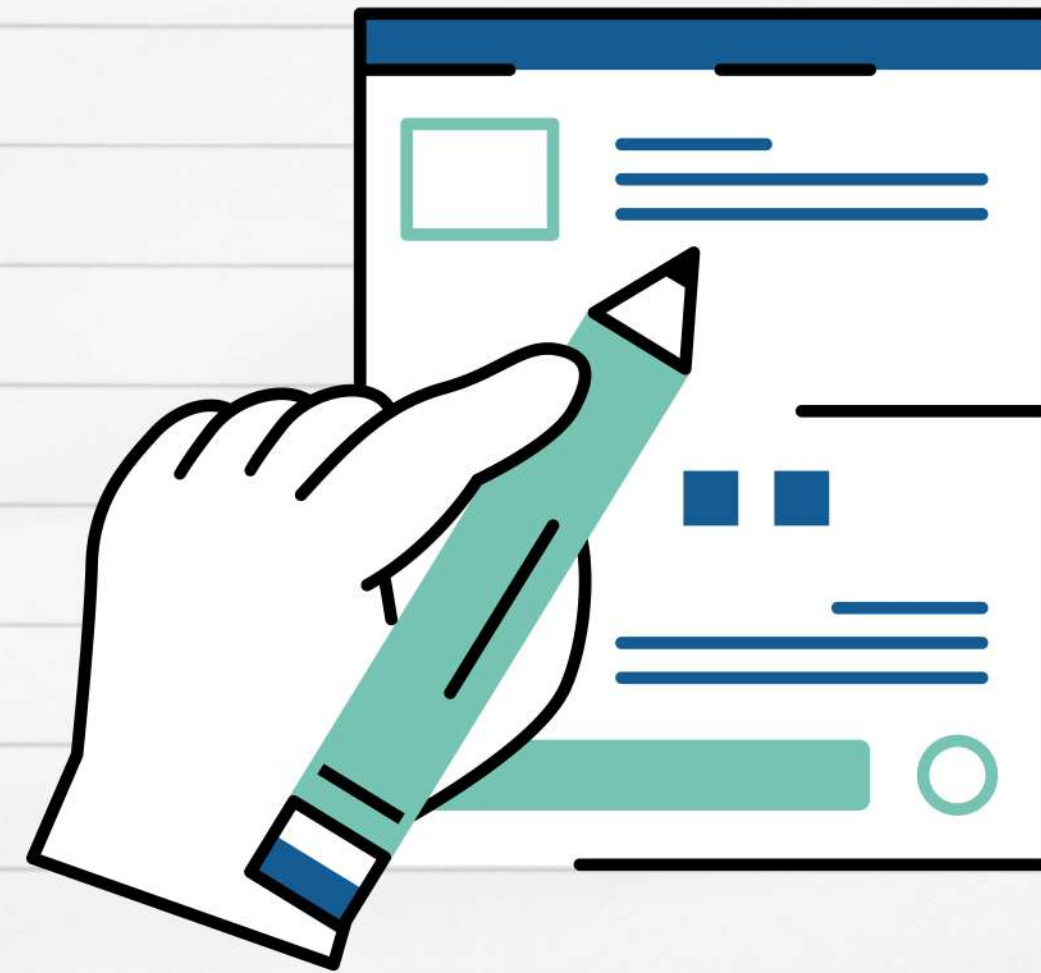




LOST AND FOUND MANAGEMENT SYSTEM

**TALHA KHURRAM
AHMAD AQEEL
WAJAHAT ULLAH KHAN**

INTRODUCTION: OVERALL PROJECT OBJECTIVE



PURPOSE:

- **STREAMLINE LOST & FOUND REPORTING IN UNIVERSITIES, OFFICES, AND PUBLIC SPACES.**
- **REDUCE MANUAL WORK AND INCREASE RECOVERY SUCCESS RATES.**

GOALS:

- **PROVIDE A USER-FRIENDLY INTERFACE FOR REPORTING AND CLAIMING ITEMS.**
- **IMPLEMENT SMART MATCHING BETWEEN LOST AND FOUND ITEMS.**
- **ENSURE SECURITY WITH AUTHENTICATION AND ADMIN VERIFICATION.**

UML DIAGRAMS

- Component Diagram
- Package Diagram
- Deployment Diagram
- Architectural Diagram
- Package Diagram

PROJECT PLANNER

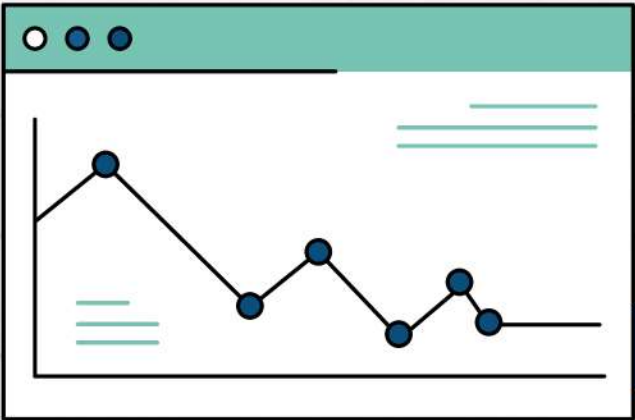
WORK BREAKDOWN

1. User Module (Auth, Profiles, Listings)
2. Item Module (Lost/Found Posting, Matching)
3. Admin Module (Verification, Dashboard)
4. Testing (Unit, Integration, E2E)

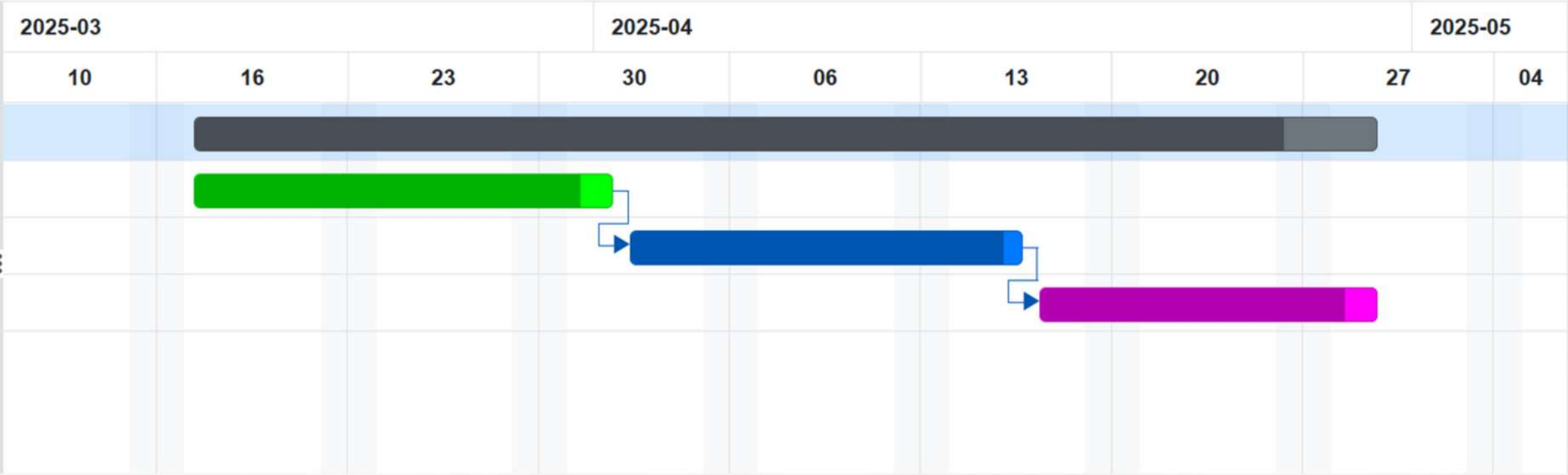
SPRINT PLANNING

1. Sprint 1: US1–US7 (Core Features)
2. Sprint 2: US8–US11 (Enhancements)
3. Sprint 3: US12–US15 (Admin & Edge Cases)

GANTT CHART



Task Name	Start	End	Duration	Progress %	Dependency
▼ Example	2025-03-17	2025-04-29	32 days	92	
Sprint 1 (MVP): User Auth, Basic Listings	2025-03-17	2025-04-01	12 days	92	
Sprint 2: Search, Claims, Notifications	2025-04-02	2025-04-16	11 days	95	2FS
Sprint 3: Admin Tools, Location Search	2025-04-17	2025-04-29	9 days	90	3FS



USER STORIES

US 1

Post Lost Item

"AS A USER, I WANT TO REPORT LOST ITEMS WITH DETAILS."

US 2

Claim a found item

"AS A USER, I WANT TO SUBMIT PROOF TO CLAIM AN ITEM."

US 3

Search with Filters

"AS A USER, I WANT TO SEARCH BY CATEGORY/LOCATION."

US 4

Location-Based Search

"AS A USER, I WANT TO SEE ITEMS NEAR ME."

US 5

Contact Finder via
Messaging

**AS A USER, CONTACT THE FINDER THROUGH I
N-APP MESSAGING**

BLACK BOX TESTING

C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\blackbox.test.js27 Feb			
Black Box Testing - User Registration & Lost Item Posting	should register a user with valid data	failed	5.017s
Black Box Testing - User Registration & Lost Item Posting	should reject registration with invalid email	failed	4.909s
Black Box Testing - User Registration & Lost Item Posting	should reject registration with missing password	passed	0.04s
Black Box Testing - User Registration & Lost Item Posting	should accept password at minimum length	failed	5.002s
Black Box Testing - User Registration & Lost Item Posting	should reject password below minimum length	failed	4.884s
Black Box Testing - User Registration & Lost Item Posting	should reject lost item post without authentication	passed	0.028s
Black Box Testing - User Registration & Lost Item Posting	should accept lost item with earliest allowed date	passed	0.033s
Black Box Testing - User Registration & Lost Item Posting	should reject lost item with date before allowed	passed	0.023s
Black Box Testing - User Registration & Lost Item Posting	should return results for valid search type	passed	0.251s
Black Box Testing - User Registration & Lost Item Posting	should handle invalid search type gracefully	passed	0.206s
Black Box Testing - Additional Cases	should reject sign in with wrong password	passed	0.212s
Black Box Testing - Additional Cases	should reject sign in with empty email	passed	0.218s
Black Box Testing - Additional Cases	should handle search with invalid category gracefully	passed	0.21s
Black Box Testing - Additional Cases	should return only one result when limit=1	passed	0.212s
Black Box Testing - Additional Cases	should return dashboard state for admin	passed	0.952s
Black Box Testing - Additional Cases	should reject lost item post with future date	passed	0.023s
Black Box Testing - Additional Cases	should reject found item post with empty description	passed	0.021s
Black Box Testing - Additional Cases	should reject lost item post with excessively long itemName	passed	0.036s
Black Box Testing - Additional Cases	should reject found item post with missing location	passed	0.029s
Black Box Testing - Additional Cases	should reject lost item post with invalid email format	passed	0.03s

1. Equivalence Class Partitioning (ECP)

Equivalence class partitioning involves dividing input data into valid and invalid partitions. Below are the applied test cases:

Test Case	Input	Class Type	Expected Output	Status
TC1	Valid email, valid password	Valid	Successful registration	✗ Failed
TC2	Invalid email (e.g., <code>abc.com</code>)	Invalid	Reject registration	✗ Failed
TC3	Empty password	Invalid	Reject registration	✓ Passed
TC4	Password = "123456" (>=6 chars)	Valid	Accept registration	✓ Passed
TC5	Password = "123" (<6 chars)	Invalid	Reject registration	✗ Failed
TC6	Auth token missing when posting	Invalid	Reject post	✓ Passed
TC7	Search type = "Electronics"	Valid	Return results	✓ Passed
TC8	Search type = "@invalid#"	Invalid	Return graceful error	✓ Passed

BLACK BOX TESTING

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2. Boundary Value Analysis (BVA)

Boundary value analysis tests the edge limits of valid ranges.

Test Case	Input	Boundary Type	Expected Output	Status
TC9	Password = 6 chars	Lower valid boundary	Accept	✗ Failed
TC10	Password = 5 chars	Upper invalid boundary	Accept	✓ Passed
TC11	Lost item date = min allowed date	Lower valid boundary	Accept post	✓ Passed
TC12	Lost item date = day before allowed	Just outside valid range	Reject post	✓ Passed
TC13	<code>limit = 1</code> in search	Edge case	Return 1 result	✓ Passed
TC14	Excessive item name length	Upper invalid boundary	Reject post	✓ Passed

BLACK BOX TESTING

C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\blackbox.test.js				27 files
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Black Box Testing - Additional Cases	should return dashboard stats for admin	passed	0.952s	
Black Box Testing - Additional Cases	should reject lost item post with future date	passed	0.023s	
Black Box Testing - Additional Cases	should reject found item post with empty description	passed	0.021s	
Black Box Testing - Additional Cases	should reject lost item post with excessively long itemName	passed	0.036s	
Black Box Testing - Additional Cases	should reject found item post with missing location	passed	0.029s	
Black Box Testing - Additional Cases	should reject lost item post with invalid email format	passed	0.03s	

3. System Functionality Validation (User Stories)

User Story	Test Case	Validation	Status
As a user, I should register with email/password	TC1-TC5	Partially met (password/email failures)	⚠️
As a user, I should post lost item only when authenticated	TC6	Met	✅
As a user, I should be able to search by category/type	TC7-TC8	Met	✅
As an admin, I should see dashboard stats	TC15	Met	✅
As a user, the system should handle bad inputs gracefully	TC8, TC10, TC14	Met	✅

WHITE BOX TESTING

TEST CASES

^ C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\app.test.js 20.698%			
User Registration & Authentication	should return 400 if email or password is missing	passed	0.171s
User Registration & Authentication	should return 200 for a new registration	passed	11.991s
User Registration & Authentication	should return 401 for invalid email on sign in	passed	0.275s
Lost & Found Item Posting	should return 401 or 404 if no token is provided when posting lost item	passed	0.036s
Lost & Found Item Posting	should return 401 or 404 if no token is provided when posting found item	passed	0.034s
Search Endpoints	should return 200 and results for search	passed	0.284s
Search Endpoints	should return 400 if lat/long are missing in nearby search	passed	0.027s
Admin Endpoints	should return 401 for invalid admin credentials	passed	0.019s
Admin Endpoints	should return analytics data for admin	passed	1.012s
Admin Endpoints	should return all users for admin	passed	0.207s
Admin Endpoints	should return all items for admin	passed	0.389s
Admin Endpoints	should return all claims for admin	passed	0.736s
Claim Controller	should return 400 or 401 if required fields are missing when creating a claim	passed	0.028s
Claim Controller	should return 400 or 401 for invalid itemId format	passed	0.03s
Claim Controller	should return 401 or 403 if not authenticated when getting finder claims	passed	0.028s
Claim Controller	should return 400, 401, or 403 for invalid claim status	passed	0.014s
Claim Controller	should return 401 or 403 if not authenticated when getting my claims	passed	0.022s
Lost Item Endpoints	should return all lost items	passed	0.058s
Lost Item Endpoints	should return 404, 401, or 403 for a non-existent lost item	passed	0.027s
Lost Item Endpoints	should return 401, 403, or 404 when deleting a lost item without auth	passed	0.017s

COVERAGE REPORT HIGHLIGHTS

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	47.02	15.97	37.83	47.1	
backend	91.48	0	66.66	91.48	
app.js	94.28	0	50	94.28	60-61
database.js	83.33	100	100	83.33	19-20
backend/config	100	100	100	100	
cloudinary.js	100	100	100	100	
backend/controllers	31.89	15.67	36.66	31.88	
adminController.js	55.84	50	46.15	55.84	...29,135-139,145-152
claimController.js	9.09	0	0	9.09	...32,141-209,218-230
foundItemPost.js	8.92	0	0	8.92	7-130
lostItemPost.js	8.47	0	0	8.47	7-143
searchItems.js	61.53	50	100	63.88	16-17,33,53,63-92
userSignUp.js	54.83	42.85	75	54.83	42-62,82,115-161
backend/middleware	52	25	50	52	
auth.js	45.45	50	100	45.45	8-14
upload.js	57.14	0	0	57.14	19-27
backend/models	100	100	100	100	
claimModel.js	100	100	100	100	
foundItemModel.js	100	100	100	100	
lostItemModel.js	100	100	100	100	
pendingUserModel.js	100	100	100	100	
userRegModel.js	100	100	100	100	
backend/routes	85.18	100	0	85.18	
adminRoutes.js	100	100	100	100	
claimRoutes.js	100	100	100	100	
lfmsRouters.js	50	100	0	50	9-12,17-20
searchRoutes.js	100	100	100	100	
userRoutes.js	100	100	100	100	



FURTHER IMPROVEMENTS

1. Mobile Apps (iOS/Android) for better accessibility.
2. AI Image Recognition for automatic item matching.
3. Blockchain for tamper-proof ownership records.
4. Multi-language Support for diverse users.
5. IoT Integration (e.g., QR tags on high-value items).

CONCLUSION



LESSONS LEARNED

Teamwork: Clear role division (user modules, admin, search) was crucial.

Testing: Early unit tests save debugging time later.

User-Centric Design: Feedback loops improved UX.

FINAL THOUGHT

"THIS SYSTEM BRIDGES THE GAP BETWEEN LOST ITEMS AND THEIR OWNERS, FOSTERING TRUST
AND EFFICIENCY IN COMMUNITIES."

**THANK
YOU VERY
MUCH!**

